HW8 – Project Planning

Team Members:

Noah Schumacher

Ryan Broyles

Manjinder Kaur

Minh Nguyen

Akshat Chaturvedi

Task List:

Task ID	Task Description	Primary Owner
T01	Research E1 - Identity Provider hosted OIDC/PKCE options and approve the provider/session approach.	Akshat Chaturvedi
T02	Design E4 - Storage schemas for VideoAsset, TrainingJob, ModelArtifact, PointCloudCache, and E2 extensions.	Akshat Chaturvedi
T03	Develop E4 - Storage SQLite persistence and checksummed binary video/model/point-cloud APIs.	Akshat Chaturvedi
T04	Specify E2 - Backend API /api/v1 contracts for reconstruction, action extraction, and training handoffs.	Minh Nguyen
T05	Develop E2 - Backend API FastAPI routing, queued workers, and failure/status handling.	Minh Nguyen
T06	Design C1 - Video Upload file-selection, metadata-validation, upload-progress, and error states.	Manjinder Kaur
T07	Develop C1 - Video Upload MP4/MOV transfer, cancellation, retry, and E4 video_id handoff.	Manjinder Kaur
T08	Develop C2 - Configuration parameter entry, validation feedback, and navigation state.	Manjinder Kaur
T09	Develop C3 - Review video/settings summary, final validation, and training-job submission.	Manjinder Kaur

T10	Develop C4 - Training Monitor progress, errors, results, and model-artifact links.	Manjinder Kaur
T11	Obtain E2 - Backend API permitted third-person demonstration clips and implement FFmpeg frame sampling.	Noah Schumacher
T12	Investigate E2 - Backend API VGGT-Omega point-cloud reconstruction on the selected clips.	Noah Schumacher
T13	Develop E2 - Backend API 3D Gaussian Splatting, D0 segmentation hook, and DNLR mesh prototype.	Noah Schumacher
T14	Investigate E2 - Backend API action-segmentation model and export timestamped ActionSegment records.	Noah Schumacher
T15	Validate E2 - Backend API reconstructed geometry and action segments, including E4 artifact references.	Noah Schumacher
T16	Develop E3 - Training & Sim Engine MuJoCo scene import and collision checks using a reference mesh.	Ryan Broyles
T17	Develop E3 - Training & Sim Engine simulation execution and robotic-action trajectory export.	Ryan Broyles
T18	Develop E3 - Training & Sim Engine diffusion-model baseline from simulated action sequences.	Ryan Broyles
T19	Test E3 - Training & Sim Engine diffusion training, evaluation, and ModelArtifact export.	Ryan Broyles
T20	Validate E3 - Training & Sim Engine integrated video-to-MuJoCo-to-model prototype and fallback behavior.	Ryan Broyles
T21	Develop E1 - Identity Provider login/callback, PKCE session checks, and protected application routes.	Akshat Chaturvedi
T22	Develop E2 - Backend API async E3/E4 job dispatch, progress persistence, and artifact retrieval.	Minh Nguyen
T23	Test E2 - Backend API integrated C1-C4 flow, authentication, retries, and failure reporting.	Minh Nguyen
T24	Test E4 - Storage checksum recovery, invalid writes, SQLite consistency, and saved-artifact downloads.	Akshat Chaturvedi
T25	Document E2 - Backend API acceptance-test evidence, deployment/runbook steps, and final SkillForge demo.	Minh Nguyen

Milestone List:

- All tasks listed; Completed Milestones highlighted in green

2A | Course deliverables: completed documents and remaining placeholders

ID	Target date	Deliverable / checkable result	How verified	Verifier
CW1	2026-09-02	Professional Biography: Team bios submitted on canvas and in repository	Check Repo and canvas assignment page	Noah Schumacher
CW2	2026-09-09	Assignment 2 - Team Formation and Project Description: Team names and project summary posted to course excel spreadsheet.	Check Repo and canvas assignment page; excel sheet	Noah Schumacher
CW3	2026-09-18	Assignment 3 - Advisor Selection and Design Constraints Essay: Fill in advisor name in spreadsheet and submit proof of advisor acceptance.	Check Repo and canvas assignment page	Noah Schumacher
CW4	2026-09-28	Assignment 4 - User Stories and Use Cases: Complete user stories and alternative paths document and submit to Repo.	Check Repo and canvas assignment page	Noah Schumacher
CW5	2026-10-02	HW5 – Design D0: D0 block/component/data-flow design exists in the provided document.	Check Repo and canvas assignment page	Noah Schumacher
CW6	2026-10-08	HW6 – Design D1: D1 details E2/E3/E4 algorithms, data model, build vs reuse, and APIs.	Check Repo and canvas assignment page	Akshat Chaturvedi
CW7	2026-10-14	HW7 – Design D2: D2 details E1/C1-C4 and adds E2 action extraction design.	Check Repo and canvas assignment page	Minh Nguyen
CW8	2026-10-28	Assignment 8 - Task List, Milestones, Timeline, and Effort Matrix: Complete doc and submit to Repo	Check Repo and canvas assignment page	Minh Nguyen
CW9	2026-10-30	Assignment 9 - Requirements Document and Technical Specification Document: Complete doc and submit to Repo	Check Repo and canvas assignment page	Noah Schumacher
CW10	2026-11-04	Assignment 10 - Test and Validation Plan Draft: Complete doc and submit to Repo	Check Repo and canvas assignment page	Akshat Chaturvedi
CW11	2026-11-11	Assignment 11 - Ethics and Responsible-Use Statement: Complete doc and submit to Repo	Check Repo and canvas assignment page	Manjinder Kaur
CW12	2026-11-18	Assignment 12 - Presentation Draft and Documented Advisor Feedback: Complete doc and submit to Repo	Check Repo and canvas assignment page	Ryan Broyles
CW13	2026-11-18	**A**ssignment 13 - Team Oral Presentation, Recorded Video, and Peer Reviews: Complete doc and submit to Repo	Check Repo and canvas assignment page	Noah Schumacher
CW14	2026-12-02	Assignment 14 - Design Report, Design Notebook, and Team and Self-Assessment: Complete doc and submit to Repo	Check Repo and canvas assignment page	Minh Nguyen
CW15	2026-12-08	Assignment #15 - Project Advisor Assessments: Complete doc and submit to Repo	Check Repo and canvas assignment page	Minh Nguyen

2B | Internal implementation milestones, every 1–2 weeks or less

ID	Date	Milestone / checkable result	Verification	Verifier
M1	2026-11-13	Foundation baseline: E1 provider recommendation, E4 entity-extension schema, and C1 upload design are ready for review.	Merge five dated decisions/artifacts, inspect one decodable MP4/MOV, and approve interface checklist.	Akshat Chaturvedi + Minh Nguyen
M2	2026-12-04	Core infrastructure proof: An upload/storage and worker baseline exists: E4 stores/retrieves a file by ID; VGGT outputs a first point cloud; MuJoCo opens a reference mesh.	Run scripted storage round-trip, inspect point count/artifact, and launch headless MuJoCo scene; record results.	Noah Schumacher + Ryan Broyles
M3	2026-12-18	Connected application foundation: C1 uploads a clip; C2 validates settings; E1 login protects an endpoint; E2 job submission responds with status.	Screen-record UI walkthrough and pass auth, upload, validation, and queued-job unit tests.	Akshat Chaturvedi + Manjinder Kaur
M4	2027-01-15	Perception and simulation prototypes: A saved geometry/mesh candidate and labeled action segments are generated; a MuJoCo trajectory and diffusion baseline run on reference data; C3 supports review.	Inspect mesh and timestamps; replay stored trajectory; run baseline training once; confirm review blocks invalid settings.	Noah Schumacher + Ryan Broyles
M5	2027-01-29	Full feature handoff: C4 reads status, E2 persists asynchronous progress/results, E4 integrity tests pass, and E3 saves a baseline model artifact.	Demonstrate queued/running/completed/failed states, checksum tests, and retrieval of a ModelArtifact record.	Minh Nguyen + Manjinder Kaur
M6	2027-02-12	End-to-end prototype acceptance: At least one permitted demonstration moves through upload, extraction/reconstruction, MuJoCo processing, baseline model production, and reviewable results, with documented fallbacks.	Execute a reproducible demo checklist; retain log, sample files, screenshots, and pass/fail record.	Noah Schumacher + Ryan Broyles
M7	2027-03-05	Release and course handoff: A tagged SkillForge prototype, Tasklist.md, planning tables, runbook, risk disposition, and demo/acceptance evidence are delivered.	Rebuild from clean checkout; run acceptance smoke test; match all 25 tasks and milestone IDs to table rows.	Minh Nguyen + Akshat Chaturvedi

Table 1 – Timeline

ID	Full task statement	Owner	Start	Finish	Depends on	Buffer %
T01	Research E1 - Identity Provider hosted OIDC/PKCE options and approve the provider/session approach. .	Akshat Chaturvedi	10/26/2026	11/06/2026	—	10
T02	Design E4 - Storage schemas for VideoAsset, TrainingJob, ModelArtifact, PointCloudCache, and E2 extensions.	Akshat Chaturvedi	10/26/2026	11/06/2026	—	15
T03	Develop E4 - Storage SQLite persistence and checksummed binary video/model/point-cloud APIs.	Akshat Chaturvedi	11/09/2026	11/20/2026	T02	20
T04	Specify E2 - Backend API /api/v1 contracts for reconstruction, action extraction, and training handoffs.	Minh Nguyen	10/26/2026	11/06/2026	—	15
T05	Develop E2 - Backend API FastAPI routing, queued workers, and failure/status handling.	Minh Nguyen	11/09/2026	11/20/2026	T02, T04	20
T06	Design C1 - Video Upload file-selection, metadata-validation, upload-progress, and error states.	Manjinder Kaur	11/02/2026	11/10/2026	—	10
T07	Develop C1 - Video Upload MP4/MOV transfer, cancellation, retry, and E4 video_id handoff.	Manjinder Kaur	11/23/2026	12/01/2026	T02, T06	20
T08	Develop C2 - Configuration parameter entry, validation feedback, and navigation state.	Manjinder Kaur	12/07/2026	12/15/2026	T04	15
T09	Develop C3 - Review video/settings summary, final validation, and training-job submission.	Manjinder Kaur	01/04/2027	01/12/2027	T05, T08	15
T10	Develop C4 - Training Monitor progress, errors, results, and model-artifact links.	Manjinder Kaur	01/18/2027	01/26/2027	T05, T09	15
T11	Obtain E2 - Backend API permitted third-person demonstration clips and implement FFmpeg frame sampling.	Noah Schumacher	11/02/2026	11/10/2026	—	15
T12	Investigate E2 - Backend API VGGT-Omega point-cloud reconstruction on the selected clips.	Noah Schumacher	11/16/2026	11/25/2026	T11	20
T13	Develop E2 - Backend API 3D Gaussian Splatting, D0 segmentation hook, and DNLR mesh-export prototype.	Noah Schumacher	12/07/2026	12/15/2026	T12	25
T14	Investigate E2 - Backend API action-segmentation model and export timestamped ActionSegment records.	Noah Schumacher	01/04/2027	01/12/2027	T11	20
T15	Validate E2 - Backend API reconstructed geometry and action segments, including E4 artifact references.	Noah Schumacher	02/01/2027	02/09/2027	T03, T13, T14	15
T16	Develop E3 - Training & Sim Engine MuJoCo scene import and collision checks using a reference mesh.	Ryan Broyles	11/09/2026	11/20/2026	T04	15
T17	Develop E3 - Training & Sim Engine simulation execution and robotic-action trajectory export.	Ryan Broyles	11/23/2026	12/01/2026	T16	20
T18	Develop E3 - Training & Sim Engine diffusion-model baseline from simulated action sequences.	Ryan Broyles	01/04/2027	01/12/2027	T17	20
T19	Test E3 - Training & Sim Engine diffusion training, evaluation, and ModelArtifact export.	Ryan Broyles	01/18/2027	01/26/2027	T18, T03	25
T20	Validate E3 - Training & Sim Engine integrated video-to-MuJoCo-to-model prototype and fallback behavior.	Ryan Broyles	03/08/2027	03/16/2027	T13, T15, T19, T22	20

ID	Full task statement	Owner	Start	Finish	Depends on	Buffer %
T21	Develop E1 - Identity Provider login/callback, PKCE session checks, and protected application routes.	Akshat Chaturvedi	11/23/2026	12/01/2026	T01	20
T22	Develop E2 - Backend API async E3/E4 job dispatch, progress persistence, and artifact retrieval.	Minh Nguyen	02/01/2027	02/09/2027	T03, T05, T17	20
T23	Test E2 - Backend API integrated C1-C4 flow, authentication, retries, and failure reporting.	Minh Nguyen	02/22/2027	03/02/2027	T07, T08, T09, T10, T21, T22	20
T24	Test E4 - Storage checksum recovery, invalid writes, SQLite consistency, and saved-artifact downloads.	Akshat Chaturvedi	12/07/2026	12/15/2026	T03	15
T25	Document E2 - Backend API acceptance-test evidence, deployment/runbook steps, and final SkillForge demo.	Minh Nguyen	03/22/2027	04/02/2027	T20, T23, T24	15

Table 1, continued | One planned-date row per milestone

Milestone	Completion date	Protected buffer preceding milestone (if major)
CW1	2026-09-02	Non-major milestone - No buffer
CW2	2026-09-09	Non-major milestone - No buffer
CW3	2026-09-18	Non-major milestone - No buffer
CW4	2026-09-28	Non-major milestone - No buffer
CW5	2026-10-02	Non-major milestone - No buffer
CW6	2026-10-08	Non-major milestone - No buffer
CW7	2026-10-14	Non-major milestone - No buffer
CW8	2026-10-28	Non-major milestone - No buffer
CW9	2026-10-30	Non-major milestone - No buffer
CW10	2026-11-04	Non-major milestone - No buffer
CW11	2026-11-11	Non-major milestone - No buffer
CW12	2026-11-18	2026-11-16 to 2026-11-17
CW13	2026-11-18	2026-11-16 to 2026-11-17
CW14	2026-12-02	2026-11-30 to 2026-12-01
CW15	2026-12-08	Non-major milestone - No buffer
M1	2026-11-13	2026-11-11 to 2026-11-12
M2	2026-12-04	2026-12-02 to 2026-12-03
M3	2026-12-18	2026-12-16 to 2026-12-17
M4	2027-01-15	2027-01-13 to 2027-01-14
M5	2027-01-29	2027-01-27 to 2027-01-28
M6	2027-02-12	2027-02-10 to 2027-02-11
M7	2027-03-05	2027-03-03 to 2027-03-04

Gantt chart | SkillForge implementation, October 2026–April 2027

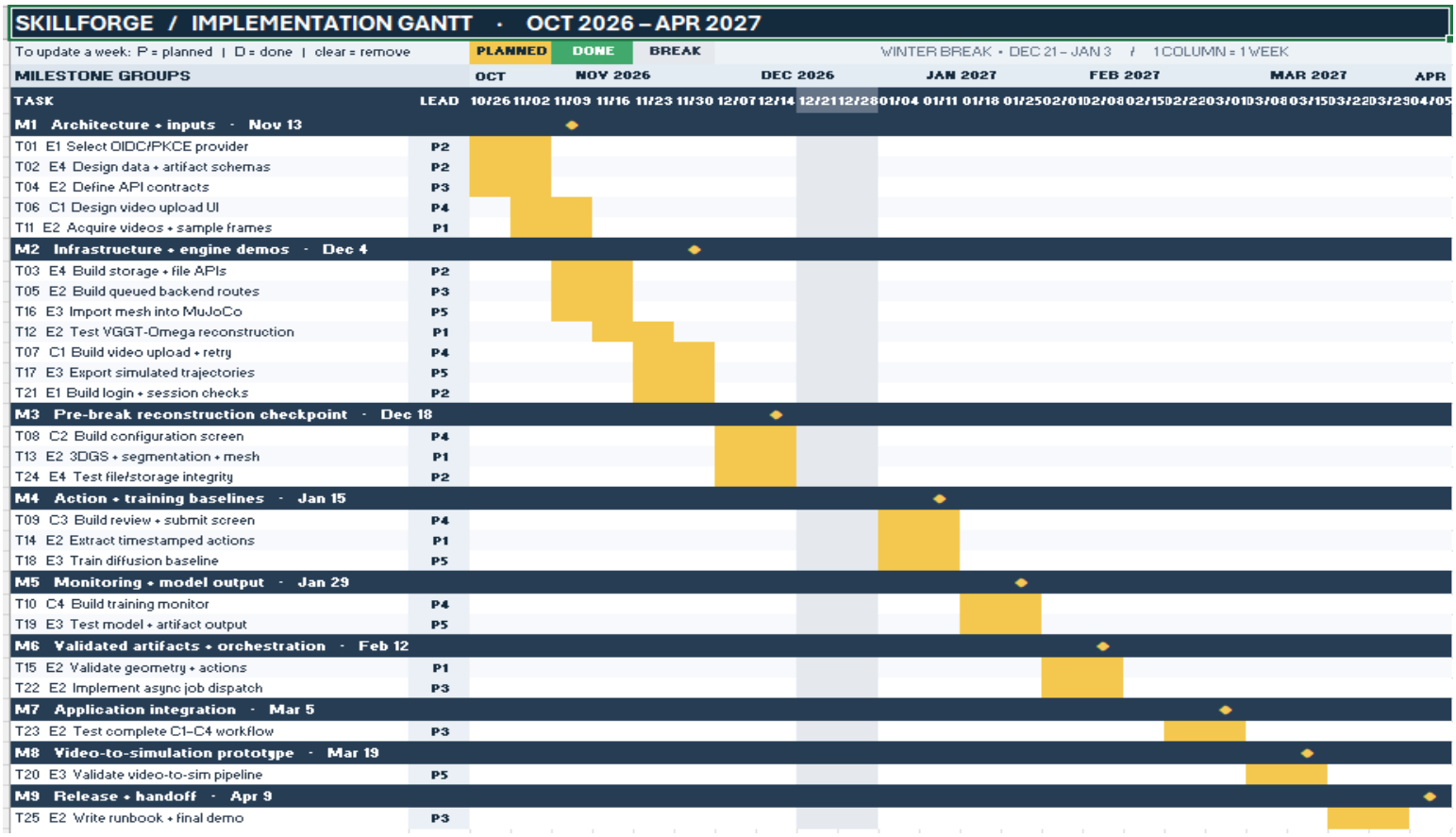


Table 2 – Effort Matrix

Asterisk (*) after buffered hours marks the primary owner; that contributor has the largest allocation for the row. Totals are projected effort, not elapsed time.

ID	Noah Schumacher	Akshat Chaturvedi	Minh Nguyen	Manjinder Kaur	Ryan Broyles	Row total	Primary owner
T01	0	8*	2	2	0	12	Akshat Chaturvedi
T02	2	12*	2	0	2	18	Akshat Chaturvedi
T03	3	15*	3	3	0	24	Akshat Chaturvedi
T04	0	3	10*	5	0	18	Minh Nguyen
T05	3	3	12*	2	4	24	Minh Nguyen
T06	0	3	0	9*	0	12	Manjinder Kaur
T07	0	4	1	19*	0	24	Manjinder Kaur
T08	0	2	2	14*	0	18	Manjinder Kaur
T09	0	2	2	14*	0	18	Manjinder Kaur
T10	0	0	4	12*	2	18	Manjinder Kaur
T11	12*	3	3	0	0	18	Noah Schumacher
T12	16*	0	4	0	4	24	Noah Schumacher
T13	21*	4	0	0	5	30	Noah Schumacher
T14	17*	0	4	0	3	24	Noah Schumacher
T15	13*	2	3	0	0	18	Noah Schumacher
T16	3	2	0	0	13*	18	Ryan Broyles
T17	4	0	3	0	17*	24	Ryan Broyles
T18	3	0	4	0	17*	24	Ryan Broyles
T19	4	2	4	0	20*	30	Ryan Broyles
T20	3	0	3	1	17*	24	Ryan Broyles
T21	0	17*	3	4	0	24	Akshat Chaturvedi
T22	0	4	11*	4	5	24	Minh Nguyen
T23	0	2	11*	9	2	24	Minh Nguyen
T24	2	13*	3	0	0	18	Akshat Chaturvedi
T25	1	0	9*	5	3	18	Minh Nguyen
TOTAL	107	101	103	103	114	528	5 tasks each

- Work breakdown is consistent enough for the resulting effort to be even across the board. In the future, misaligned work effort will be re-examined closer to the performed date.

Work Breakdown Structure

1. Phase 1 — Architecture and foundations

- 1.T01: Research E1 - Identity Provider hosted OIDC/PKCE options and approve the provider/session approach. Owner: Akshat Chaturvedi.
- 1.T02: Design E4 - Storage schemas for VideoAsset, TrainingJob, ModelArtifact, PointCloudCache, and E2 extensions. Owner: Akshat Chaturvedi.
- 1.T04: Specify E2 - Backend API /api/v1 contracts for reconstruction, action extraction, and training handoffs. Owner: Minh Nguyen.

2. Phase 2 — Application, identity, and persistence

- 2.T03: Develop E4 - Storage SQLite persistence and checksummed binary video/model/point-cloud APIs. Owner: Akshat Chaturvedi.
- 2.T05: Develop E2 - Backend API FastAPI routing, queued workers, and failure/status handling. Owner: Minh Nguyen.
- 2.T06: Design C1 - Video Upload file-selection, metadata-validation, upload-progress, and error states. Owner: Manjinder Kaur.
- 2.T07: Develop C1 - Video Upload MP4/MOV transfer, cancellation, retry, and E4 video_id handoff. Owner: Manjinder Kaur.
- 2.T08: Develop C2 - Configuration parameter entry, validation feedback, and navigation state. Owner: Manjinder Kaur.
- 2.T09: Develop C3 - Review video/settings summary, final validation, and training-job submission. Owner: Manjinder Kaur.
- 2.T10: Develop C4 - Training Monitor progress, errors, results, and model-artifact links. Owner: Manjinder Kaur.
- 2.T21: Develop E1 - Identity Provider login/callback, PKCE session checks, and protected application routes. Owner: Akshat Chaturvedi.
- 2.T24: Test E4 - Storage checksum recovery, invalid writes, SQLite consistency, and saved-artifact downloads. Owner: Akshat Chaturvedi.

3. Phase 3 — Video perception and reconstruction

- 3.T11: Obtain E2 - Backend API permitted third-person demonstration clips and implement FFmpeg frame sampling. Owner: Noah Schumacher.
- 3.T12: Investigate E2 - Backend API VGGT-Omega point-cloud reconstruction on the selected clips. Owner: Noah Schumacher.
- 3.T13: Develop E2 - Backend API 3D Gaussian Splatting, D0 segmentation hook, and DNLR mesh-export prototype. Owner: Noah Schumacher.
- 3.T14: Investigate E2 - Backend API action-segmentation model and export timestamped ActionSegment records. Owner: Noah Schumacher.
- 3.T15: Validate E2 - Backend API reconstructed geometry and action segments, including E4 artifact references. Owner: Noah Schumacher.

4. Phase 4 — Simulation and policy baseline

- 4.T16: Develop E3 - Training & Sim Engine MuJoCo scene import and collision checks using a reference mesh. Owner: Ryan Broyles.

- 4.T17: Develop E3 - Training & Sim Engine simulation execution and robotic-action trajectory export. Owner: Ryan Broyles.
- 4.T18: Develop E3 - Training & Sim Engine diffusion-model baseline from simulated action sequences. Owner: Ryan Broyles.
- 4.T19: Test E3 - Training & Sim Engine diffusion training, evaluation, and ModelArtifact export. Owner: Ryan Broyles.
- 4.T20: Validate E3 - Training & Sim Engine integrated video-to-MuJoCo-to-model prototype and fallback behavior. Owner: Ryan Broyles.

5. Phase 5 — System integration and release

- 5.T22: Develop E2 - Backend API async E3/E4 job dispatch, progress persistence, and artifact retrieval. Owner: Minh Nguyen.
- 5.T23: Test E2 - Backend API integrated C1-C4 flow, authentication, retries, and failure reporting. Owner: Minh Nguyen.
- 5.T25: Document E2 - Backend API acceptance-test evidence, deployment/runbook steps, and final SkillForge demo. Owner: Minh Nguyen.

Risk Register

ID	Category	L	I	Strategy	Risk / consequence	Mitigation	Owner	Trigger
R1	Technical	High	High	Reduce	Dynamic third-person video may yield sparse/unstable VGGT, 3DGS, or low-coverage scenes.	Use short high-overlap clips, quality thresholds, and a preapproved static/reference clip fallback; checkpoint T12 and T13 before B2.	Noah Schumacher	Sparse reconstruction or low-confidence point cloud on the approved clip by Nov 25.
R2	Technical	Medium	High	Reduce	DNLR/segmentation geometry may not be collision-safe or physically scaled.	Run mesh topology/scale checks; use simplified collision proxy and reference mesh until validated; isolate visual mesh from collider.	Ryan Broyles	Mesh still fails manifold/scale/collision checks by Dec 15.
R3	Technical	High	High	Reduce	Extracted human action segments may not map to executable robot trajectories or a useful policy.	Prototype synthetic/reference actions first, mark untrusted inferred mappings explicitly, constrain demo task to one action, define baseline success/failure metrics.	Ryan Broyles	No replayable action trajectory or meaningful segment-to-action path by Jan 12.
R4	External	Medium	High	Reduce	GPU availability, model weights, library licenses, or cloud cost may block reconstruction/training.	Check rights and hardware early; cache dependencies, cap frame count, set budget limit, and use a CPU/lightweight precomputed artifact fallback.	Noah Schumacher	Required weights unavailable, out-of-memory errors, or cloud budget exceeded by Nov 25.
R5	External	Medium	Medium	Reduce	Hosted OIDC service outage or callback misconfiguration interrupts protected workflow.	Approve provider in T01; use mock identity only for local development and re-run actual OIDC integration before demo.	Akshat Chaturvedi	OIDC login cannot pass a protected-route test by Dec 1.

ID	Category	L	I	Strategy	Risk / consequence	Mitigation	Owner	Trigger
R6	Team	Medium	High	Reduce	Five members may face uneven workloads, absence, or interface handoff delays.	Review hour totals weekly, enforce merged contract by T04, pair-review critical PRs, use protected gate buffers, reassign support hours promptly.	Minh Nguyen	Owner task slips more than two workdays or a member is unavailable for three.
R7	Team/technical	High	High	Avoid / reduce	D1 reconstruction, D2 action extraction, and E4 entities/APIs may disagree on ownership or status/ID semantics.	Freeze E2/E3/E4 contract and schema extensions in T02/T04; use contract tests; block unreviewed incompatible API changes.	Minh Nguyen	Incompatible E2/E3/E4 fields or state transitions detected at contract test by Nov 20 or Feb 9.
R8	Technical	Medium	High	Reduce	Large video upload or local storage writes may fail, lose data, or leave orphan metadata.	Stream to temporary file, SHA-256, transactional metadata writes, cleanup/retry tests, monitor free disk.	Akshat Chaturvedi	Corrupt file, orphan row, checksum failure, or storage below 20% free by Dec 15.
R9	Technical/schedule	High	High	Reduce	Full diffusion policy learning from exocentric video is unlikely to be robust within seven weeks.	Limit acceptance to one reproducible baseline on simulated trajectories; label full end-to-end robotics transfer as stretch; protect M6 demo.	Ryan Broyles	Diffusion baseline cannot train/evaluate or replay by Jan 26.
R10	External	Medium	Medium	Avoid	Public video copyright/consent or incompatibility prevents legal, reproducible demo use.	Choose licensed/publicly permissible or internally captured clips; record provenance, codec, and scene criteria.	Noah Schumacher	Demo clip permissions or provenance cannot be documented by Nov 10.